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Econ 50: Using Big Data to Solve Economic and Social Problems

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Dual Markets: Upward Mobility in Destin and Okaloosa County

About Destin

Destin is a small beach town and fishing village located in Okaloosa County on the Florida Panhandle. Known for its white sand and bright teal water, it has become a major tourism destination for Southern Americans, with its population rising from around 14,000 to 28,000–42,000 in the summer ("Florida Circumnavigational Saltwater Paddling Trail," Segment 2). Okaloosa County is the home for residents of the Crestview-Fort Walton Beach-Destin metropolitan area, with much of the tourism being concentrated in Destin, while the remaining metropolitan areas house tourist-dependent service workers.

Pre-1970, Destin was a very small fishing village, with only around 1,000 permanent residents. The economy was mostly centered around fishing and some small amount of tourism. However, in the early 1970s, the first high rise condominiums in Destin were built (*East Pass Inlet Management Study*). In fact, Census data shows that the share and number of seasonal and recreational housing units in Florida increased sharply between 1970 and 1980, rising from roughly 3–4% of housing units to about 7% (Figure 1, "Historical Census of Housing Tables"). Because Destin was becoming a major coastal tourist destination around this time, it's likely that much of this statewide increase in seasonal housing was especially concentrated in areas like Okaloosa County. As a result, tourism became the primary driver for economic development, which reshaped the local economy from one of fishing to hospitality, real estate, and seasonal services ("Florida Circumnavigational Saltwater Paddling Trail," Segment 2).

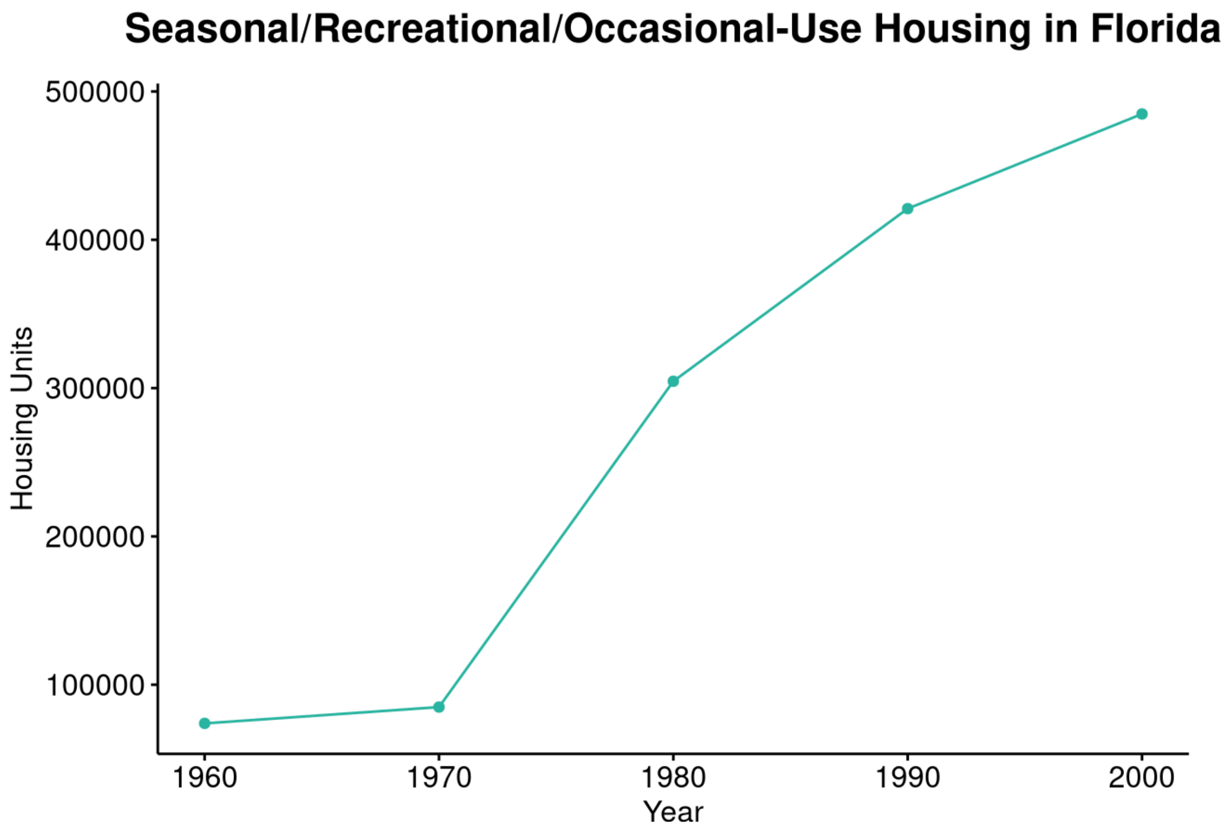

Figure 1: Seasonal/Recreational/Occasional-Use Housing in Florida.

Source: Census Bureau, Historical Census of Housing Tables: Vacation Homes (1960-2000).

Hypothesis

As a result of the tourism market's expansion in Destin and surrounding areas, the area's housing market was split into two:

- one focused on tourism/coastal markets, which brought in vacation rentals, part-time residents, and outside money; and
- one focused on the local labor market, of which its members serviced the tourists via

hotels, restaurants, etc. and thus had lower wages and lived in more permanent residencies.

We believe that the economic mobility inequality across Destin was likely a result of this housing and job market split. More specifically, the development of beachfront condos and vacation properties in the 70s and 80s created areas with relatively few permanent low-income families (or, at the very least, a positively selected subset of low-income residents in terms of stability, education, or economic prospects), while other parts of Destin housed the lower-income service workers tied to the tourism industry.

Evaluation

We use data from the Opportunity Atlas, along with ZIP-level social capital measures from Opportunity Insights to analyze this claim. However, our datasets do not contain any specific measure for the amount of seasonal housing that a region might have. Homeownership itself is a noisy measure, as high amounts of seasonal rentals may imply a region has low homeownership, but high homeownership also signals wealth, which is likely correlated with upward mobility. Because we cannot directly observe these tourism-driven housing patterns, we instead move to testing the implications of our hypothesis by examining the characteristics of the neighborhoods where lower-income service workers are likely to live. We assume that the workers who service the tourism industry are more likely to live in regions with higher poverty, lower incomes, and lower educational attainment. Additionally, service workers are likely to live in areas with lower exposure to high-income individuals, meaning those areas will have lower economic connectedness. As such, having high economic connectedness might be itself an indicator for whether a region is high-mobility.

Based on the exploration of the above datasets, we determined that a few particular variables would be of interest, those being the ones described in Table A. For each model, we ran a linear regression across multiple subsets of the data. In particular, when possible, we looked at the same linear regression for residents living in Destin, Okaloosa County, the Pensacola Commuting Zone (CZ), Florida, and the entirety of the United States. When data for Destin in specific was not available or would lack variability across predictors (as Destin only contains six Census tracts), we attempted to evaluate against the Okaloosa County and the Pensacola CZ subsets instead. This is because Okaloosa County represents the smallest administrative region (besides ZIP Code, of which Destin only has one of) that captures Destin, while the Pensacola CZ would (hopefully) capture the labor market structure best, as individuals may travel across city and county lines to commute to work. We then compared these results to that of Florida and the US in general, who will serve as baselines for interpreting our results.

Variable Name	Kind	Nickname	Description
`kfr_pooled_pooled_p25`	Outcome	Mobility	Pooling all racial/ethnic groups and genders, the Absolute Mobility at the 25th Percentile using Household Income as Income Concept
`lockbox_kfr_pooled_pooled_p25`	Outcome	LockboxMobility	Pooling all racial/ethnic groups and genders, the Absolute Mobility at the 25th Percentile using Household Income as Income Concept
`poor_share2000`	Predictor	Poverty	Share of individuals in the tract below the federal poverty line, measured in the decennial Census of the year 2000.
`hhinc_mean2000`	Predictor	Income	Mean Household Income 2000 (2015 dollars). Obtained from 2000 Decennial Census.

frac_coll_plus 2000	Predictor	Education	Fraction of Residents w/ a College Degree or More in 2000.
jobs_total_5mi _2015	Predictor	Jobs	Number of Primary Jobs within 5 Miles in 2015.
jobs_highpay_5 mi_2015	Predictor	HighPayJobs	Number of High-Paying (>USD40,000 annually) Jobs within 5 Miles in 2015 (number of jobs in LEHD within 5 miles of tract centroid paying $3,333 / month or more).
ec_zip	Predictor	Connectedness	Economic Connectedness.
ec_std	Predictor	StdConnectedn ess	Standard deviation of Economic Connectedness.

Table A: Variables Used in Evaluation.

Model 1: Spatial Sorting

As stated before, we believe that the tourism industry has sorted service workers into worse-off areas. If this is true, we should observe that tracts with higher poverty, lower income, and lower educational attainment have lower upward mobility. To test this prediction, I estimate a regression of upward mobility on poverty, mean household income, and educational attainment across all the described subsets. More formally, this can be described by the equation

$$\text{Mobility}_i = \beta_0 + \beta_1 \cdot \text{Poverty}_i + \beta_2 \cdot \text{Income}_i + \beta_3 \cdot \text{Education}_i + \epsilon_i$$

using the variables found in Table A.

Firstly, there were not enough observations or variation in the key regressors at the Destin level (as predicted), leaving some undefined coefficient estimates. As such, we look towards the remaining subsets and continue our regression analysis.

Across Okaloosa, the Pensacola CZ, Florida, and the US, we find that our poverty coefficient is always negative, our income coefficient is always positive, and our education coefficient is always positive. In Okaloosa, a 10 percentage point increase in the share of poverty is associated with a around a 3.6 percentile drop in mobility. In the Pensacola CZ, this percentile drop is only around 2.2. A $10,000 increase in mean household income is associated with approximately a .40 percentile increase in mobility in Okaloosa. For the Pensacola CZ, this value is .62. And, in Okaloosa, a 10 percentage point increase in the share of residents with a college degree or more is associated with approximately a 1.09 percentile increase in mobility. In the Pensacola CZ, this effect is around 0.76 percentile points.

All of these results were statistically significant at the $\alpha = 0.05$ level. Importantly, these findings are consistent with the idea of some kind of spatial sorting mechanism, in which lower-income, less-education populations are concentrated in areas with lower levels of upward mobility. Additionally, this spatial sorting seems to hold true for not just Okaloosa and the Pensacola CZ, but for Florida and the US as a whole. But, this relationship seems to hold the strongest in Okaloosa County, suggesting the mechanism may be particularly relevant in the local context around Destin. The full data for these can be found in Table 1.

Model 2a: Job Density and Pay

If access to economic opportunity drives upward mobility as predicted, we should observe that areas with greater access to jobs, especially high-paying ones, have higher levels of upward mobility. In particular, if low-mobility areas in Destin or Okaloosa are disadvantaged because they are physically or economically separated from high-paying jobs, then job density and access to high-paying jobs should be strong predictors of mobility. To test this, I estimate a regression of mobility on measures of nearby job density and access to high-paying jobs, the equation of which can be described as:

$$\text{Mobility}_i = \beta_0 + \beta_1 \cdot \text{Jobs}_i + \beta_2 \cdot \text{HighPayJobs}_i + \epsilon_i$$

The results show that job density and access to high-paying jobs are not strong predictors of upward mobility in any region. Across Okaloosa and the Pensacola CZ, the results are inconsistently signed and not statistically significant. However, the results for Florida and the US are actually statistically significant. But, the coefficients are incredibly small and still retain inconsistent signs. As such, this weak relationship between job access and mobility casts doubt on a purely "job access" explanation and motivates a closer look into social and network-based mechanisms in the next model. The full data for this model can be found in Table 2.

Model 2b/2c: Economic Connectedness

If mobility depends not only on the presence of jobs but also on access to opportunity networks, then areas with higher economic connectedness should have higher upward mobility. In the case of Destin and Okaloosa County, this tests whether residents in low-mobility areas are disconnected from higher-income networks, even if they happen to live near a tourism-driven labor market. We estimate:

$$\text{Mobility}_i = \beta_0 + \beta_1 \cdot \text{Connectedness}_i + \epsilon_i$$

We also realized that a single unit of economic connectedness is not well comparable across the 4 regions because a 1-unit increase isn't entirely realistic or interpretable. Additionally, the range of `ec_zip` varies across each region (e.g., the minimum and maximum values of `ec_zip` in Okaloosa might be 0.4 and 0.8, while in the US it might be 0.2 and 0.9). Thus, we will standardize `ec_zip` to have a mean of 0 and a standard deviation of 1, and then re-run the regressions to get more comparable coefficients across the regions. We estimate:

$$\text{Mobility}_i = \beta_0 + \beta_1 \cdot \text{StdConnectedness}_i + \epsilon_i$$

Our results show that for both models, economic connectedness is positively associated with upward mobility in most samples. In Okaloosa, a single increase in the standard deviation of economic connectedness is associated with a 3.15 percentile increase in upward mobility. In contrast, the same increase in the standard deviation of economic connectedness is associated with a 1.4 percentile increase in Florida and a 1.8 percentile increase in the US. The Pensacola CZ estimate is positive but not statistically significant.

Unlike job density, economic connectedness appears to be a meaningful predictor of upward mobility. This suggests that opportunity is not simply about whether jobs are nearby, but whether residents have access to higher-income social networks that may transmit information, expectations, referrals, or other advantages. The full data for Model 2b and 2c are available in Tables 3 and 4, respectively.

Model 3: Economic Connectedness, Controlling for Poverty, Income, and Education

While Model 2b/2c were useful, they did not yet show whether connectedness matters independently of poverty, income, and education. If in the case that they don't, then connectedness may simply have been proxying for spatial sorting (or some other mechanism). We estimate:

$$\text{Mobility}_i = \beta_0 + \beta_1 \cdot \text{StdConnectedness}_i + \beta_2 \cdot \text{Poverty}_i + \beta_3 \cdot \text{Income}_i + \beta_4 \cdot \text{Education}_i + \epsilon_i$$

Results show that economic connectedness remains positively associated (although less strongly) with economic mobility in Okaloosa, Florida, and the national sample, even after controlling for poverty, income, and education. In Okaloosa, a single increase in the standard deviation of economic connectedness is now associated with a 2.01 percentile increase in upward mobility. For the Pensacola CZ, the economic connectedness coefficient is near 0 and negative, but remains statistically insignificant. Thus, this supports the idea that service-worker communities may be disadvantaged not only via lower income or lower education levels, but also because they are less connected to higher-income networks, and that the spatial sorting in Destin contributed to this limited access to said networks. The full data for these results can be found in Table 5.

Model 4: Economic Connectedness, Controlling for Poverty, Income, and Education (Lockbox)

Because many of our judgments on the inclusion of predictors were conducted during exploratory analysis of the Opportunity Atlas dataset, it would be wise to double check that these judgements and predictions hold up against unseen data. We additionally protect against potential noise or data snooping in our non-lockboxed dataset. If our findings still hold true, we have even stronger evidence of the claims made in Model 3. In particular, we would like to estimate:

$$\text{LockboxMobility}_i = \beta_0 + \beta_1 \cdot \text{StdConnectedness}_i + \beta_2 \cdot \text{Poverty}_i + \beta_3 \cdot \text{Income}_i + \beta_4 \cdot \text{Education}_i + \epsilon_i$$

Again, results largely replicate the original findings. Economic connectedness remains positive and statistically significant in Okaloosa County, Florida, and the national sample. The coefficient is smaller than in the original sample, but the sign and significance are consistent. As before, the Pensacola commuting zone does not show a statistically significant connectedness effect. The full results for these data can be found in Table 6.

Table 1: Model 1: Spatial Sorting

	Destin	Okaloosa	Pensacola CZ	Florida	US
Intercept	49.46	41.48	36.13	37.54	40.22
	(0.011)	(< 0.001)	(< 0.001)	(< 0.001)	(< 0.001)
poor_share2000	-74.01	-36.06	-22.33	-20.15	-21.08
	(0.711)	(< 0.001)	(0.000006)	(< 0.001)	(< 0.001)
hhinc_mean2000	NA	0.000040	0.000062	0.000030	0.000041
	NA	(0.00037)	(0.026)	(< 0.001)	(< 0.001)
frac_coll_plus2000	NA	10.94	7.58	14.42	8.48
	NA	(< 0.001)	(0.034)	(< 0.001)	(< 0.001)
R^2	0.038	0.530	0.613	0.468	0.385

Table 2: Model 2a: Job Access

	Destin	Okaloosa	Pensacola CZ	Florida	US
Intercept	19.42	43.29	41.31	39.85	43.00
	(0.189)	(< 0.001)	(< 0.001)	(< 0.001)	(< 0.001)
jobs_total_5mi	0.00317	0.0000115	0.0000469	0.0000771	-0.00000825
	(0.188)	(0.684)	(0.684)	(< 0.001)	(< 0.001)
jobs_highpay_5mi	-0.00619	0.0000611	-0.000361	-0.000167	0.0000133
	(0.439)	(0.241)	(0.293)	(< 0.001)	(< 0.001)
R^2	0.651	0.132	0.189	0.026	0.001

Table 3: Model 2c: Economic Connectedness (Standardized)

	Okaloosa	Pensacola CZ	Florida	US
Intercept	44.89	39.11	41.46	40.56
	(< 0.001)	(< 0.001)	(< 0.001)	(< 0.001)
ec_std	3.15	0.85	1.39	1.83
	(0.0000001)	(0.164)	(< 0.001)	(< 0.001)
R^2	0.250	0.032	0.092	0.118

Table 4: Model 2c: Economic Connectedness (Standardized)

	Okaloosa	Pensacola CZ	Florida	US
Intercept	44.89	39.11	41.46	40.56
	(< 0.001)	(< 0.001)	(< 0.001)	(< 0.001)
ec_std	3.15	0.85	1.39	1.83
	(0.0000001)	(0.164)	(< 0.001)	(< 0.001)
R^2	0.250	0.032	0.092	0.118

Table 5: Model 3: Connectedness with Controls

	Okaloosa	Pensacola CZ	Florida	US
Intercept	47.22	45.11	38.48	37.89
	(< 0.001)	(< 0.001)	(< 0.001)	(< 0.001)
ec_std	2.01	-0.04	0.39	0.47
	(0.00011)	(0.931)	(0.0016)	(< 0.001)
(< 0.001)				
poor_share2000	-37.71	-41.22	-19.61	-21.55
	(0.00021)	(0.00002)	(< 0.001)	(< 0.001)
hhinc_mean2000	-0.000032	0.000030	0.000046	0.000064
	(0.226)	(0.644)	(< 0.001)	(< 0.001)
frac_coll_plus2000	13.11	-4.14	7.56	5.55
	(0.025)	(0.688)	(< 0.001)	(< 0.001)
R^2	0.517	0.514	0.507	0.590

Table 6: Model 4: Lockbox Validation

	Okaloosa	Pensacola CZ	Florida	US
Intercept	53.00	57.78	39.88	42.78
	(< 0.001)	(< 0.001)	(< 0.001)	(< 0.001)
ec_std	1.64	0.11	0.34	0.36
	(0.00115)	(0.817)	(0.0016)	(< 0.001)
poor_share2000	-45.03	-52.84	-14.40	-24.74
	(0.000009)	(0.0000005)	(< 0.001)	(< 0.001)
hhinc_mean2000	-0.000050	-0.000091	0.000051	0.000036
	(0.053)	(0.195)	(< 0.001)	(< 0.001)
frac_coll_plus2000	2.57	-15.68	-0.61	-0.28
	(0.649)	(0.153)	(0.662)	(0.371)
R^2	0.315	0.387	0.389	0.502

Interpretation

Given these results, we believe economic opportunity in the Destin area is not only shaped by who lives in a neighborhood, but also by networks these residents are connected to.

First, it was consistently shown that socioeconomic characteristics (poverty, education, income) were all strong predictors of upward mobility. Areas with higher poverty and lower income experienced significantly lower mobility. This supports our idea of spatial sorting, where low-income service workers tied to the tourism industry generally are concentrated in neighborhoods with fewer economic opportunities.

Secondly, and arguably more importantly, economic connectedness, even after controlling for the above socioeconomic characteristics, presented itself as a strong predictor for upward mobility. Access to higher-income networks seem to play a much more important role in economic outcomes than one may realize, and beyond traditional measures of neighborhood disadvantage.

By contrast, job access did not appear to explain upward mobility. Simply having enough jobs nearby did not seem sufficient to generate much upward mobility. As such, policymakers should focus on interventions that increase interaction between different socioeconomic groups, as opposed to simply adding jobs (even high-paying ones) to the labor market. Mixed-income housing (of which Destin lacks much of), school integration (of which Okaloosa has seemingly made worse), and/or community programs could be useful and important in improving mobility.

Limitations

Arguably, the biggest limitation is that Destin itself is too small to analyze directly. Attempts to analyze Destin showed results with little to no variation, and thus traditional regression analysis became nearly impossible. Our results relied moreso on Okaloosa County and the Pensacola CZ. Additionally, the Pensacola CZ often showed contradictory and statistically insignificant results compared to Okaloosa. There may be some heterogeneity across the region that simply isn’t present in the smaller Okaloosa County population.

Furthermore, vacation rental activity was not directly measured here, and much of our results are only able to explain whether the implications of our hypothesis hold rather than the hypothesis itself. This data was, of course, not present in the Opportunities Atlas dataset, but even more concerning is the possible lack of this data at such a granular level like the Census tracts at all.

Finally, we cannot definitively conclude that economic connectedness causes higher mobility. The analysis we conducted was purely observational, and there are likely omitted variables (e.g. school quality, family structure, local institutions) that connectedness is simply a proxy for, as opposed to being the root cause.

Next Steps

To further establish causality, further work could conduct a difference-in-differences approach to determine whether some shock to Destin contributed to upward mobility. For instance, the changing of zoning laws to permit resort/condo/short-term-rental development in specific regions of Destin might be able to help us better confirm whether, indeed, the presence and construction of seasonal housing positively influences upward mobility by means of introducing wealthier networks to the area. This would help us confirm our hypothesis directly, as opposed to confirming the implications of the hypothesis as we have done today.

Additionally, we could use a regression discontinuity design approach to determine whether some geographical aspect of Destin is influencing upward mobility. For instance, the Opportunity Atlas shows worse economic activity in areas south of Highway 98 (the major highway in Destin that divides access between inhabitants from the Gulf of Mexico, and where most tourism-oriented housing is often located) as opposed to north (Figure 2). If the areas are otherwise similar, any discontinuity in mobility or economic connectedness at the boundary could be interpreted as evidence of the causal effect of tourism-driven development.

Of course, both of these approaches require more fine-grained data about the upward mobility that may simply not exist yet at this small of a scale. However, at this scale, information becomes more and more revealing about the individuals in the dataset, which limits how much information we can know about Destin. Ultimately, better data collection that can preserve privacy well is needed to validate either of these approaches and the hypothesis itself.

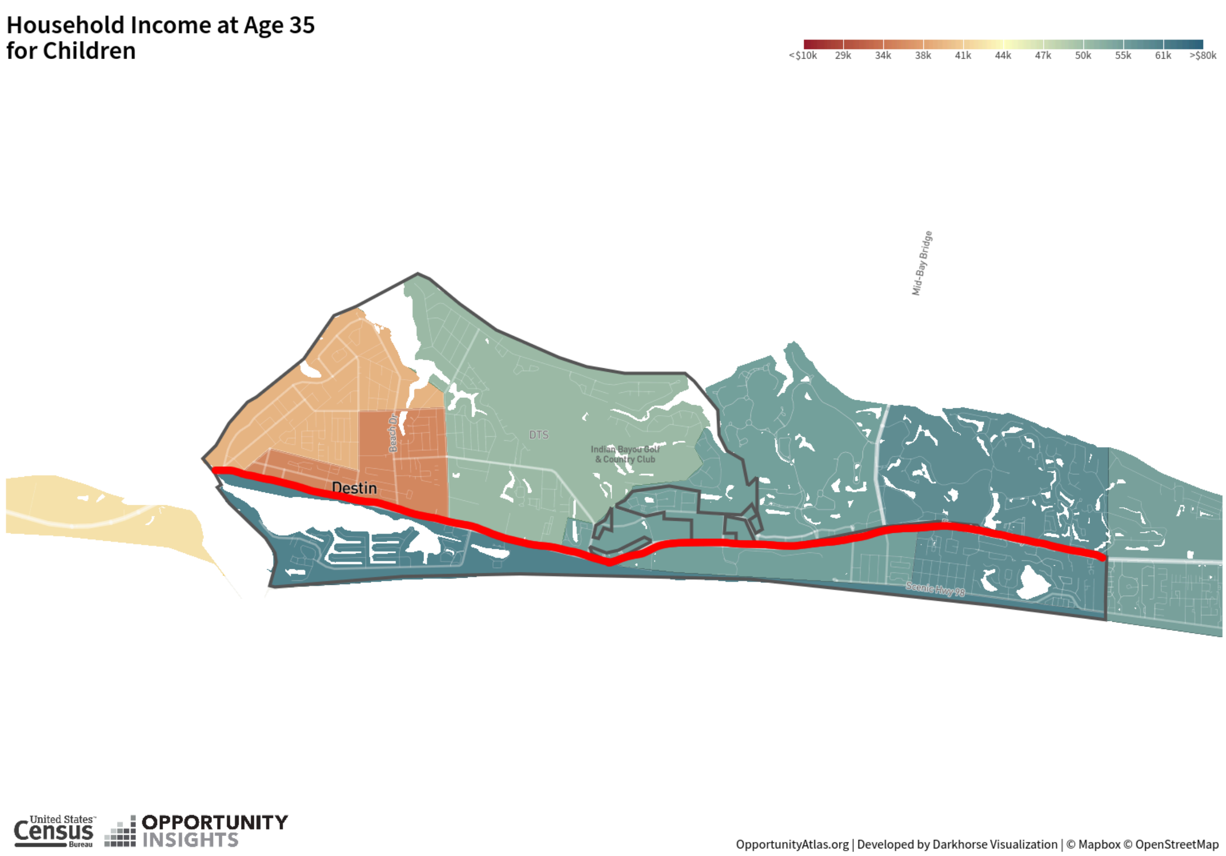

Figure 2: Household Income at Age 35 for Children, with Highway 98 highlighted in red and Destin bordered in gray.

Source: Opportunity Atlas

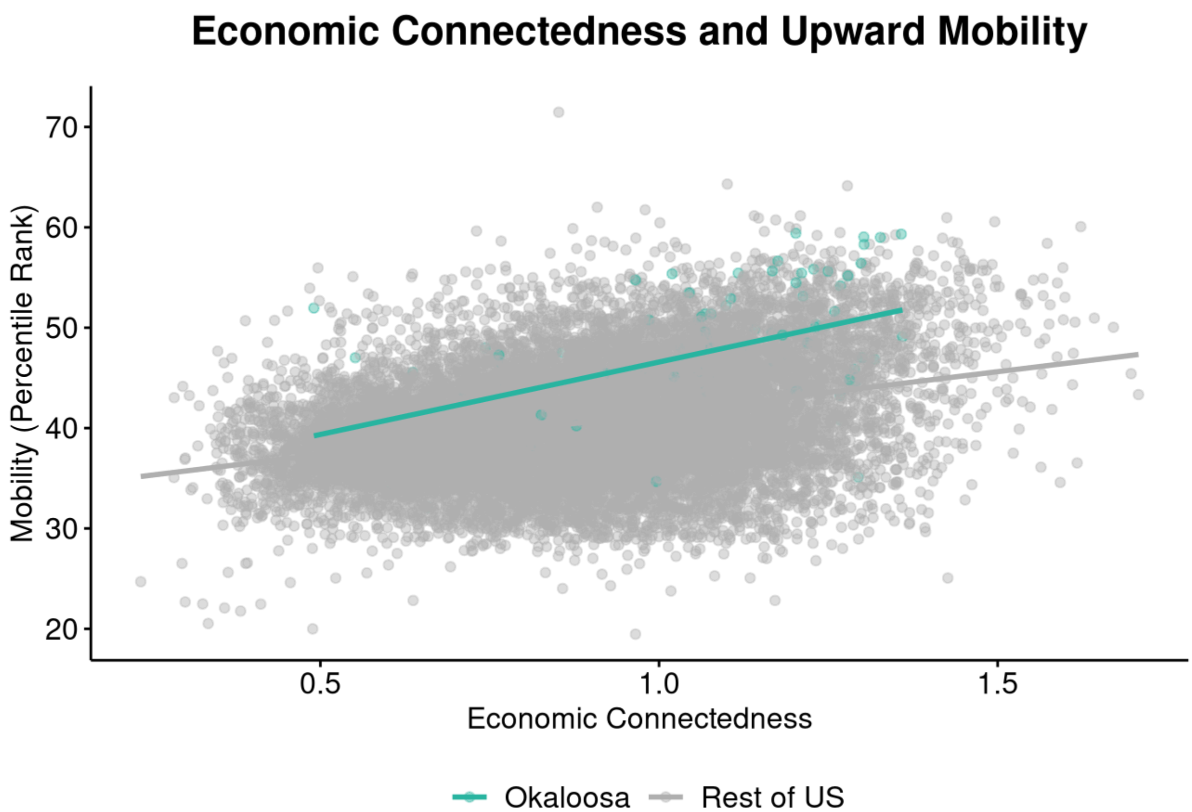

Figure 3: Economic Connectedness and Upward Mobility

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